

TM1621 LCD Driver

Titan Micro Electronics - Datasheet V1.4 (English translation of the Chinese original)

Translator's note: text is translated as written in the source; obvious errors in the original are flagged in [brackets]. Figures and timing diagrams are described in text rather than reproduced.

Overview

The TM1621 is a memory-mapped, multi-function LCD driver. Its software-configurable features make it suitable for a wide range of LCD applications, including LCD modules and display subsystems. Only 4 or 5 pins are needed to connect the host controller to the TM1621. The TM1621 also has a power-saving command to reduce system power consumption.

Features

- Operating voltage: 2.4 ~ 5.2 V
- Built-in 256 kHz RC oscillator
- Optional external 32 kHz crystal or 256 kHz clock source input
- Selectable 1/2 or 1/3 bias and 1/2, 1/3 or 1/4 duty
- On-chip time-base frequency source
- Buzzer with two selectable frequencies
- Power-saving command to reduce power consumption
- Built-in time-base generator and watchdog timer (WDT)
- Time-base or WDT overflow output
- Eight time-base / WDT clock sources
- 32x4 LCD driver
- Built-in 32x4-bit display RAM
- Four-wire serial interface
- On-chip LCD drive frequency source
- Software-configurable
- Data-mode and command-mode instructions
- Three data access modes
- VLCD pin provided for adjusting the LCD operating voltage
- Packages: SSOP48, LQFP44

Block Diagram

The device consists of: a Control and Timing circuit (pins OSCO, OSCI, /CS, /RD, /WR, DATA, VDD, VSS); Display RAM; LCD Driver / Bias circuit (outputs COM0~COM3, SEG0~SEG31, input VLCD); Tone Frequency Generator (BZ, /BZ); and Watchdog Timer and Time Base Generator (/IRQ).

/CS: chip select. BZ, /BZ: sound output. /WR, /RD, DATA: serial interface. COM0~COM3, SEG0~SEG31: LCD outputs. /IRQ: time-base or WDT overflow output.

Pin Assignment

SSOP48 (pin: name): 1-7: SEG7..SEG1; 8: SEG0; 9: /CS; 10: /RD; 11: /WR; 12: DATA; 13: VSS; 14: OSCO; 15: OSCI; 16: VLCD; 17: VDD; 18: /IRQ; 19: BZ; 20: /BZ; 21-24: COM0..COM3; 25-48: SEG31..SEG8 (pin 25 = SEG31, pin 48 = SEG8).

LQFP44 (pin: name): 1: /CS; 2: /WR; 3: DATA; 4: VSS; 5: VLCD; 6: VDD; 7: BZ; 8: /BZ; 9-12: COM0..COM3; 13-32: SEG31..SEG12 (pin 13 = SEG31, pin 32 = SEG12); 33: SEG11; 34-44: SEG10..SEG0. The LQFP44 has no /RD, OSCI, OSCO or /IRQ pins.

Pin Function Definitions

Symbol	Name	SSOP48	LQFP44	Description
/CS	Chip select	9	1	When /CS is high, reads/writes of data and commands are invalid and the serial interface circuit is reset. When /CS is low (used as input), reads/writes of data and commands are valid.
/RD	Data input	10	/	On the falling edge of /RD, data from TM1621 memory is output onto the DATA line; the host can latch it on the next rising edge.
/WR	Data input	11	2	On the rising edge of /WR, data on the DATA line is written into the TM1621.
DATA	Serial data I/O	12	3	Serial data logic input/output, with external pull-up resistor.
VSS	Logic ground	13	4	System ground.
OSCI, OSCO	External crystal	14/15	/	A 32.768 kHz crystal connected between OSCI and OSCO generates the system clock. If another external clock source is used, connect it to OSCI. If the on-chip RC oscillator is used, OSCI and OSCO must be left floating.
VLCD	LCD supply input	16	5	LCD power supply input.
VDD	Logic positive	17	6	System positive supply.
/IRQ	Time base / watchdog	18	/	Time-base or watchdog timer overflow flag, NMOS open-drain output.
BZ, /BZ	Tone output	19/20	7/8	Sound frequency output.
COM0~CO M3	Common outputs	21~24	9~12	LCD common drive outputs.
SEG0~SEG 31	Segment outputs	8~1 / 48~25	44~13	LCD segment drive outputs.

Absolute Maximum Ratings

Supply voltage: -0.3 V ~ 0.5 V [as printed in the original; probably intended to be 5.5 V]. Storage temperature: -50 ~ 125 °C.

Input voltage: VSS-0.3 V ~ VDD+0.3 V. Operating temperature: -25 ~ 75 °C.

DC Electrical Characteristics

Symbo l	Description	VDD	Condition	Min	Typ	Max	Unit
VDD	Operating voltage			2.4		5.2	V
IDD	Operating current	3V	No load, on-chip RC oscillator		150	300	uA
		5V			300	600	uA
IDD	Operating current	3V	No load, crystal		60	120	uA
		5V			120	240	uA
IDD	Operating current	3V	No load, external clock source		100	200	uA
		5V			200	400	uA
ISTB	Standby current	3V	No load, power-saving mode		0.1	5	uA
		5V			0.3	10	uA
VIL	Input low voltage	3V	DATA, /WR, /CS, /RD	0		0.6	V
		5V		0		1.0	V
VIH	Input high voltage	3V	DATA, /WR, /CS, /RD	2.4		3.0	V
		5V		4.0		5.0	V
IOL1	DATA, BZ, /BZ, /IRQ sink current	3V	VOL=0.3V	0.5	1.2		mA
		5V	VOL=0.5V	1.3	2.6		mA
IOH1	DATA, BZ, /BZ source current	3V	VOH=2.7V	-0.4	-0.8		mA
		5V	VOH=4.5V	-0.9	-1.8		mA

Symbo l	Description	VDD	Condition	Min	Typ	Max	Unit
IOL2	LCD common sink current	3V	VOL=0.3V	80	150		uA
		5V	VOL=0.5V	150	250		uA
IOH2	LCD common source current	3V	VOH=2.7V	-80	-120		uA
		5V	VOH=4.5V	-120	-200		uA
IOL3	LCD segment sink current	3V	VOL=0.3V	60	120		uA
		5V	VOL=0.5V	120	200		uA
IOH3	LCD segment source current	3V	VOH=2.7V	-40	-70		uA
		5V	VOH=4.5V	-70	-100		uA
RPH	Pull-up resistance	3V	DATA, /WR, /CS, /RD	40	80	150	kOhm
		5V		30	60	100	kOhm

AC Electrical Characteristics

Symbo l	Description	VDD	Condition	Min	Typ	Max	Unit
fSYS1	System clock	3V / 5V	On-chip RC oscillator		256		kHz
fSYS2	System clock	3V / 5V	Crystal		32.768		kHz
fSYS3	System clock	3V / 5V	External clock source		256		kHz
fLCD	LCD clock		On-chip RC oscillator		fSYS1/1024		Hz
			Crystal		fSYS2/128		Hz
			External clock source		fSYS3/1024		Hz
tCOM	COM clock period (n = number of COMs)				n / fLCD		s
fCLK	Serial data clock	3V				300	kHz
		5V				500	kHz
fTONE	Tone frequency		On-chip RC oscillator		2.0 or 4.0		kHz
tDD	DATA pin output delay	3V				2	us
		5V				1	us
tCS	Serial interface reset pulse width		/CS	4			us

Application Circuit

The host controller (MCU) and the TM1621 form a display system. MCU connections: /CS, /RD (optional, marked *), /WR and DATA go to the TM1621; /IRQ returns to the MCU through a pull-up resistor R to VDD. The OSCI/OSCO pins are wired for one of: external clock 1 / external clock 2 (from the MCU clock output), on-chip oscillator (pins floating), or a 32768 Hz crystal with capacitors C1 and C2. VDD feeds VLCD through a variable resistor VR. BZ and /BZ drive a piezo buzzer. COM0~COM3 and SEG0~SEG31 drive the LCD panel (1/2 or 1/3 bias; 1/2, 1/3 or 1/4 duty).

Notes:

- VLCD must be lower than VDD.
- VR adjusts the VLCD output voltage. For VDD = 5 V and VLCD = 4 V, VR = 15 kOhm +/-20%.
- R is an external pull-up resistor to VDD, R = 10 kOhm +/-20%.
- C1, C2 are external crystal capacitors. Recommended values: crystal tolerance +/-10 ppm: 0~10 pF; 10~20 ppm: 10~20 pF.

System Architecture

1. Display Memory (RAM)

The static display memory (RAM) stores the displayed data in a 32x4-bit format. RAM data is mapped directly to the LCD driver and can be accessed with the READ, WRITE and READ-MODIFY-WRITE commands. Each

address (0~31) corresponds to one segment (SEG0~SEG31); the four bits Bit3..Bit0 correspond to COM3..COM0.

2. System Oscillator

The TM1621 system clock is used to generate the time-base/watchdog timer (WDT) clock frequency, the LCD drive clock and the tone frequency. The clock source can be the on-chip RC oscillator (256 kHz), a crystal (32.768 kHz) or an external 256 kHz clock selected by software. The SYS DIS command stops the system clock and the LCD bias generator; it applies only to the on-chip RC oscillator or crystal. When the system clock is stopped, the LCD display is blank and the time-base/WDT functions are also disabled.

The LCD OFF command turns off the LCD bias generator; after that, SYS DIS can be used to reduce system power, in which case SYS DIS is the power-saving command. If the system clock is an external clock, SYS DIS cannot turn off the system clock or reduce power. In crystal mode, a 32 kHz signal source may also be connected to the OSC1 pin, but then the system cannot enter power-saving mode. In external 256 kHz clock mode the system likewise cannot enter power-saving mode. At power-on the TM1621 is in the SYS DIS state.

3. Time Base and Watchdog Timer (WDT)

The time-base generator is an 8-stage up counter that produces an accurate time base. The WDT consists of the time-base generator and a 2-stage up counter, and can generate an interrupt when the host controller or another subsystem is in an abnormal state. When the WDT overflows, an on-chip WDT overflow flag is set; a command option routes the time-base generator or the WDT overflow flag to the /IRQ pin. There are 8 time-base/WDT clock frequencies: $f_{WDT} = 32 \text{ kHz} / 2^n$, where $n = 0 \sim 7$ is selected by command. The 32 kHz is the system frequency, derived from the crystal (32.768 kHz), the on-chip oscillator (256 kHz) or the external 256 kHz clock.

If the system source is the on-chip oscillator (256 kHz) or the external (256 kHz) clock, it is divided down to 32 kHz by a 3-stage prescaler. The time-base generator and the WDT share the same 8-stage counter, so commands related to them must be used with care. For example, the WDT DIS command disables the time-base generator, and the WDT EN command not only enables the time-base generator but also enables output of the WDT overflow flag (connected to the /IRQ pin). After TIMER EN is executed, the WDT is not connected to /IRQ and the output of the time-base generator is connected to the /IRQ pin.

The CLR WDT command clears the WDT overflow flag; the time-base generator value can be cleared with CLR WDT or CLR TIMER. CLR WDT or CLR TIMER should be executed before the corresponding WDT EN or TIMER EN command; before executing /IRQ EN, CLR WDT or CLR TIMER should be executed first; and CLR TIMER must be executed before switching from WDT mode to time-base mode. When the WDT overflows, the /IRQ pin stays low until CLR WDT or /IRQ DIS is executed. When /IRQ output is disabled, the /IRQ pin is high impedance. /IRQ EN / /IRQ DIS enable or disable the /IRQ output; /IRQ EN routes the time-base generator or WDT overflow flag to the /IRQ pin. When using the on-chip oscillator or a crystal, the oscillator can be turned on or off with the related system commands, which reduces system power. In power-saving mode the time base/WDT is disabled.

If an external clock is selected as the system frequency source, the SYS DIS command has no effect and the system cannot enter power-saving mode. The TM1621 keeps operating until system power is turned off or the external clock is removed. After power-on, the /IRQ output is disabled.

4. Tone Output

The TM1621 contains a simple tone generator that outputs a pair of drive signals on the BZ and /BZ pins to produce a single tone. The TONE 4K and TONE 2K commands select one of two frequencies (4 kHz and 2 kHz). TONE ON and TONE OFF turn the tone output on and off. BZ and /BZ are the tone output pins and form a differential drive pair for driving a buzzer. When the system is disabled or tone output is disabled, BZ and /BZ stay low.

5. LCD Driver

The TM1621 is a 128-dot (32x4) LCD driver. It can be configured by software for 1/2 or 1/3 LCD bias and 2, 3 or 4 commons, which makes it suitable for many LCD applications. The LCD drive clock is derived by dividing the system clock; the LCD drive clock frequency is kept at 256 Hz, whether generated from the 32.768 kHz crystal, the on-chip RC oscillator or the external clock. The LCD-related commands are shown below.

Name	Command code	Function
LCD OFF	100 0000 0010 X	Turn off LCD output

Name	Command code	Function
LCD ON	100 0000 0011 X	Turn on LCD output
BIAS & COM	100 0010 abXc X	c=0: 1/2 bias; c=1: 1/3 bias ab=00: 2 commons; ab=01: 3 commons; ab=10: 4 commons

The leading bold digits '100' denote the command-mode type. When commands are executed consecutively, the mode type code of all commands except the first is ignored. LCD OFF disables the LCD bias generator and thus turns the display off; LCD ON enables the bias generator and turns the display on. BIAS&COM is the LCD-module-related command, which makes the TM1621 compatible with most LCD modules.

6. Command Format

The TM1621 is software-configurable. Two modes of commands configure the TM1621 and transfer the data shown on the LCD. The configuration mode of the TM1621 is called command mode and has type code 100. Command mode includes a system configuration command, a system frequency selection command, an LCD configuration command, a tone frequency selection command, a timer/WDT setting command and an operation command. Data mode includes the READ, WRITE and READ-MODIFY-WRITE operations. The table below lists the data and command mode type codes.

Operation	Mode	Type code
READ	Data	110
WRITE	Data	101
READ-MODIFY-WRITE	Data	101
COMMAND	Command	100

The mode command should be executed before data or command transfer; if consecutive commands are executed, the command-mode code 100 is ignored. When the system is in non-consecutive command mode or non-consecutive-address data mode, /CS should be set to '1' and the previous operating mode is reset. When /CS returns to '0', the new operating-mode type code must be run first.

7. Interface

The TM1621 uses only four pins for its interface. /CS initializes the serial interface circuit and terminates communication between the host controller and the TM1621. When /CS is '1', data and commands between the host and the TM1621 are invalid and the interface is initialized. Before issuing a mode command or changing mode, the serial interface of the TM1621 must be initialized with a high-level pulse. DATA is the serial data input/output pin; reading/writing data and writing commands are all done through DATA. /RD is the read clock input: on the falling edge of /RD, data is output on DATA, and the host should read the data between the rising edge of /RD and the next falling edge. /WR is the write clock input: on the rising edge of /WR, the data, address and commands on DATA are written into the TM1621. The optional /IRQ pin can be used as an interface between the host and the TM1621; by software setting, /IRQ outputs the timer output or the WDT overflow flag. Once /IRQ is connected to the host, the time-base or WDT function can be realized.

8. Timing Diagrams

(Diagrams not reproduced; the sequences are described below. The host should read DATA between a rising edge of /RD and the next falling edge.)

READ mode (command code 110): with /CS low, the host sends on DATA (clocked by /WR): 1 1 0, then address bits A5~A0. The TM1621 then outputs data bits D0~D3 on DATA, clocked by /RD. For another address, /CS is pulsed high and the sequence repeats (Memory Address 1 / Data, Memory Address 2 / Data).

READ mode (consecutive addresses): after one 1 1 0 + A5~A0 sequence, /RD clocks out Data(MA), Data(MA+1), Data(MA+2), Data(MA+3), ... with the address auto-incrementing.

WRITE mode (command code 101): /WR clocks in 1 0 1, address A5~A0, then data D0~D3. /CS is pulsed high between separate address/data groups.

WRITE mode (consecutive addresses): after 1 0 1 + A5~A0, successive 4-bit data nibbles Data(MA), Data(MA+1), Data(MA+2), ... are written to consecutive addresses.

READ-MODIFY-WRITE mode (command code 101): /WR clocks in 1 0 1 and the address; data is read out on /RD, then new data is written for the same address (Data(MA1) read, then Data(MA1) written), and so on for further

addresses with /CS pulsed between them.

READ-MODIFY-WRITE mode (consecutive addresses): after one 1 0 1 + A5~A0 sequence, the read/write pairs Data(MA), Data(MA+1), Data(MA+2), ... are performed on consecutive addresses.

Command mode (command code 100): /WR clocks in 1 0 0, then 9-bit commands C8~C0 (Command 1, Command ..., Command i) back-to-back, followed by /CS high and then a new command or data mode.

Data and command modes: each /CS-low period starts with a mode code (command or data), followed by address and data.

Command Summary

Command	Code	D/C	Function	Default at power-on
READ	110 a5a4a3a2a1a0 d0d1d2d3	D	Read RAM data	
WRITE	101 a5a4a3a2a1a0 d0d1d2d3	D	Write data to RAM	
READ-MODIFY-WRITE	101 a5a4a3a2a1a0 d0d1d2d3	D	Read and write data	
SYS DIS	100 0000 0000 X	C	Turn off system oscillator and LCD bias generator	Y
SYS EN	100 0000 0001 X	C	Turn on system oscillator	
LCD OFF	100 0000 0010 X	C	Turn off LCD bias generator	Y
LCD ON	100 0000 0011 X	C	Turn on LCD bias generator	
TIMER DIS	100 0000 0100 X	C	Disable time-base output	
WDT DIS	100 0000 0101 X	C	Disable WDT overflow flag output	
TIMER EN	100 0000 0110 X	C	Enable time-base output	
WDT EN	100 0000 0111 X	C	Enable WDT overflow flag output	
TONE OFF	100 0000 1000 X	C	Turn off tone output	Y
TONE ON	100 0000 1001 X	C	Turn on tone output	
CLR TIMER	100 0000 11XX X	C	Clear time-base generator	
CLR WDT	100 0000 111X X	C	Clear WDT state	
XTAL 32K	100 0001 01XX X	C	System clock source: crystal	
RC 256K	100 0001 10XX X	C	System clock source: on-chip RC oscillator	Y
EXT 256K	100 0001 11XX X	C	System clock source: external clock	
BIAS 1/2	100 0010 abX0 X	C	LCD 1/2 bias option; ab=00: 2 commons, 01: 3 commons, 10: 4 commons	
BIAS 1/3	100 0010 abX1 X	C	LCD 1/3 bias option; ab=00: 2 commons, 01: 3 commons, 10: 4 commons	
TONE 4K	100 010X XXXX X	C	Tone frequency 4 kHz	
TONE 2K	100 011X XXXX X	C	Tone frequency 2 kHz	
/IRQ DIS	100 100X 0XXX X	C	Disable /IRQ output	Y
/IRQ EN	100 100X 1XXX X	C	Enable /IRQ output	
F1	100 101X X000 X	C	Time-base/WDT clock output 1 Hz	
F2	100 101X X001 X	C	Time-base/WDT clock output 2 Hz	
F4	100 101X X010 X	C	Time-base/WDT clock output 4 Hz	
F8	100 101X X011 X	C	Time-base/WDT clock output 8 Hz	
F16	100 101X X100 X	C	Time-base/WDT clock output 16 Hz	
F32	100 101X X101 X	C	Time-base/WDT clock output 32 Hz	
F64	100 101X X110 X	C	Time-base/WDT clock output 64 Hz	
F128	100 101X X111 X	C	Time-base/WDT clock output 128 Hz	Y
TOPT	100 1110 0000 X	C	Test mode	
TNORMAL	100 1110 0011 X	C	Normal mode	Y

Note: X = 0 or 1; a5~a0 = RAM address; d3~d0 = RAM data. The code column is shown with the 3-bit mode prefix (100) followed by the 9-bit command; spaces are added for readability only and the grouping differs slightly from the original table's print layout.

D/C: data/command mode

All the bold digits 110, 101 and 100 are mode command codes. 100 is the command-mode type code; when consecutive commands are run, the mode type code of every command except the first is ignored. The tone frequency source and the time-base/WDT clock source come from the on-chip 256 kHz RC oscillator, the 32.768 kHz crystal or the external 256 kHz clock; the frequency calculation is as described earlier. It is recommended that, after power-on reset, the host controller initializes the TM1621, because if power-on reset fails the TM1621 will not work properly.

Pin Drive Waveforms

The original figure shows the waveforms for displaying the digit '5' on a seven-segment (plus decimal point) digit with 1/4 duty and 1/3 bias drive. Segments a, b, c, d, f, g, h (middle bar) and dp are wired between COM0~COM3 and SEG0/SEG1. Waveforms are shown for COM0~COM3, SEG0 and SEG1 over one frame, with levels:

- V3 = VDD (VDD being the LCD supply voltage)
- V2 = 2/3 VDD
- V1 = 1/3 VDD
- V0 = 0

Bare Die Pad Diagram and Coordinates

Die size: 1510 x 1740 μm ; pad size 90 x 90 μm . Pads 1~8 run along the left edge (bottom-left corner is pad 7/8 at the bottom), pads 8~17 along the bottom, pads 18~30 up the right edge, pads 31~41 along the top, and pads 42~48 down the left edge. Note: the substrate is connected to low level (GND).

Pad	Name	X (μm)	Y (μm)	Remark
1	SEG7	117	813.2	
2	SEG6	117	698.2	
3	SEG5	117	583.2	
4	SEG4	117	468.2	
5	SEG3	117	353.2	
6	SEG2	117	238.2	
7	SEG1	117	117	
8	SEG0	232	117	
9	CSN (/CS)	347	117	
10	RDN (/RD)	462	117	
11	WRN (/WR)	577	117	
12	DATA	692	117	
13	GND	818	117	Connect to substrate
14	OSCO	933	117	
15	OSCI	1048	117	
16	VLCD	1163	117	
17	VDD	1278	87	
18	IRQN (/IRQ)	1393	105.6	
19	BZ	1393	235.2	
20	BZN (/BZ)	1393	350.2	
21	COM0	1393	468.2	
22	COM1	1393	583.2	
23	COM2	1393	698.2	
24	COM3	1393	813.2	
25	SEG31	1393	928.2	
26	SEG30	1393	1043.2	
27	SEG29	1393	1158.2	
28	SEG28	1393	1273.2	

Pad	Name	X (um)	Y (um)	Remark
29	SEG27	1393	1388.2	
30	SEG26	1393	1503.2	
31	SEG25	1393	1623	
32	SEG24	1272.5	1623	
33	SEG23	1157.5	1623	
34	SEG22	1042.5	1623	
35	SEG21	927.5	1623	
36	SEG20	812.5	1623	
37	SEG19	697.5	1623	
38	SEG18	582.5	1623	
39	SEG17	467.5	1623	
40	SEG16	352.5	1623	
41	SEG15	237.5	1623	
42	SEG14	117	1623	
43	SEG13	117	1503.2	
44	SEG12	117	1388.2	
45	SEG11	117	1273.2	
46	SEG10	117	1158.2	
47	SEG9	117	1043.2	
48	SEG8	117	928.2	

Package Outline Drawings

SSOP48 (dimensions in mm)

Symbol	Min	Max	Symbol	Min	Max
A	15.77	15.97	C3	0.2	0.4
A1	0.20	0.35	C4	0.12	0.25
A2	0.635 TYP		D	1.41 TYP	
A3	0.5 TYP		D1	0.61	0.91
A4	10.2 TYP		h	0.381	0.635
B	10.01	10.61	phi1	2.2 TYP	
B1	7.39	7.59	theta1	15 deg TYP	
B2	8.6 TYP		theta2	15 deg TYP	
C	2.41	2.78	theta3	4 deg TYP	
C1	2.18	2.38	theta4	8 deg TYP	
C2	1.067 TYP				

LQFP44 (common dimensions, units = millimeters)

Symbol	Min	Nom	Max
A	-	-	1.60
A1	0.05	-	0.15
A2	1.35	1.40	1.45
A3	0.59	0.64	0.69
b	0.31	-	0.44
b1	0.30	0.35	0.40
c	0.13	-	0.18
c1	0.12	0.127	0.134
D	11.80	12.00	12.20
D1	9.90	10.00	10.10
E	11.80	12.00	12.20

Symbol	Min	Nom	Max
E1	9.90	10.00	10.10
e	0.70	0.80	0.90
L	0.45	0.60	0.75
L1	1.00 REF		
L2	0.25 BSC		
R1	0.08	-	-
R2	0.08	-	0.20
S	0.20	-	-
theta	0 deg	3.5 deg	7 deg
theta1	0 deg	-	-
theta2	11 deg	12 deg	13 deg
theta3	11 deg	12 deg	13 deg

All specs and applications shown above are subject to change without prior notice by Titanmec. (The circuits and specifications above are for reference only; if the company makes corrections, no separate notice will be given.)